Identifying Skin Cancer Using AI

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Abstract

Skin cancer stands as one of the leading three cancers in Australia, underscoring the critical importance of early detection and diagnosis. Despite the existence of procedures like monthly self-examinations and screening, a considerable number of individuals face challenges in recognizing and assessing effective methods for identifying potential skin cancers. While recent advancements in smartphone apps utilising AI present promising prospects for skin cancer identification, their accuracy remains uncertain owing to limited research. The significance of this research lies in its capacity to transform the detection of skin cancer, facilitating early diagnosis and treatment interventions. Through the utilisation of AI technology, individuals can promptly detect suspicious skin lesions, resulting in enhanced prognosis and outcomes. This study aims to narrow the divide between the escalating incidence of skin cancer and the demand for accessible, precise diagnostic instruments.

Introduction

Skin cancer is one of the most prevalent diseases in the world, especially in Australia where “at least 2 in 3 Australians will be diagnosed with skin cancer in their lifetime” (Cancer Council, 2023). Early and accurate diagnoses are crucial to increasing survival rates and improving outcomes. Traditional diagnosis methods include self-examination, which can be inaccurate and difficult for inexperienced individuals, and monthly dermatologists' inspection/ screening, which can be inaccessible and time-consuming. Recently, Artificial Intelligence (AI) has been developed as an automatic detection method for skin cancer, offering a more accurate, cheaper and faster diagnosis. While these tools do show promise, it is unclear if the AI system is clinically reliable and accurate there is not enough research.

Related Work

A. A Deep Learning System for Differential Diagnosis of Skin Diseases

The article "A deep learning system for differential diagnosis of skin diseases" in Nature Medicine details a DLS developed using 16,114 de-identified teledermatology cases to diagnose 26 common skin conditions and provide secondary predictions for 419 conditions. The DLS matched dermatologists' diagnostic accuracy and outperformed PCPs and NPs. In a validation set of 963 cases, it achieved a top-1 accuracy of 66% and top-3 accuracy of 90%, with consistent performance across skin tones and demographics. The DLS's use of multiple images and clinical data enhances its accuracy, supporting non-specialist clinicians and potentially improving access to dermatological care. Despite limitations, the DLS shows promise for enhancing diagnostic accuracy and efficiency in teledermatology and primary care, with future work aimed at broader validation and clinical impact assessment.

B. AI-Powered Diagnosis of Skin Cancer: A contemporary Review, Open Challenges and Future Research Directions

The article "AI-Powered Diagnosis of Skin Cancer: A Contemporary Review, Open Challenges and Future Research Directions" provides a comprehensive overview of machine learning and deep learning techniques used in diagnosing skin cancer, stressing the importance of early detection for effective treatment. It addresses challenges in skin cancer diagnosis due to the scarcity of specialists and advocates for AI-driven automated diagnosis systems. The survey compares various datasets and reviews on automated skin cancer diagnosis, discussing insights and future research directions, including classification of skin cancer types, stages of automated dermoscopy image analysis, and the importance of AI methods like deep learning. Furthermore, it outlines the methodology used for literature review and emphasises the need for machine learning and deep learning models in enhancing early detection and prognosis. The article explores machine learning and deep learning techniques for diagnosing skin cancer, emphasising early detection's importance. It discusses challenges like specialist scarcity and advocates for AI-driven diagnosis systems. It compares datasets and reviews, highlighting future research directions. Various machine learning methods, including neural networks and ensemble learning, are analysed for their efficacy, strengths, and limitations. Deep learning techniques like recurrent neural networks and convolutional neural networks are examined, with CNNs identified as the most versatile. Open challenges such as communication barriers and dataset limitations are discussed, along with proposed future research directions like combining AI with next-gen sequencing and implementing teledermatology. Overall, the article underscores AI's potential in healthcare while acknowledging the need to address ethical, legal, and technical challenges for successful implementation.

C. Skin Cancer Classification with Deep Learning: A Systematic Review

The article provides a comprehensive exploration of skin cancer classification using deep learning, highlighting the significance of early detection. It addresses challenges such as imbalanced datasets, cross-domain adaptability, and model robustness. Various types of dermatological images and popular datasets used in skin cancer research are discussed, along with the successes and limitations of convolutional neural networks (CNNs). Solutions to frontier challenges such as data imbalance and model efficiency are presented, alongside a comparison of commonly used datasets. The article offers valuable insights for researchers aiming to advance deep learning methods in skin cancer diagnosis.

Additionally, the article discusses challenges and future research directions in integrating AI and deep learning techniques in skin cancer diagnosis. It addresses the communication barrier between dermatologists and AI, dataset availability and features, patient perspectives on AI, variation in lesion images, dermatological image acquisition, and ethical and legal perspectives. Future research directions include combining AI with genomic sequencing, developing AI-powered decision support systems, utilising smart robotics and wearable computing, and leveraging teledermatology. Overall, the article underscores the importance of addressing technical, ethical, and practical challenges to advance AI-driven skin cancer diagnosis and improve patient outcomes.

Problem Formulation

Traditional AI-based methods are extremely accurate in a variety of classification tasks, including multi-label and binary classification, however, these same machine-learning algorithms are ineffective for complex diagnoses such as skin cancer. Conventional AI-based skin classification methods mainly rely on extracting and classifying features from dermatoscopic pictures. Due to the wide variety of skin lesions, these computerised identifications frequently result in misdiagnoses. For example, some existing models cannot identify the difference between benign and malignant cancer cells. This is because different skin lesions can have identical features or the same lesions can have different features, and it can be challenging to classify them into different categories. Moreover, existing training datasets lack a diversity of skin types and conditions, which can result in a biased algorithm that, if universally applied, often misdiagnoses. Additionally, the model needs to be trained with many high-resolution images to produce an effective system, resulting in the algorithm requiring more expensive processing power and training time. Hence, our research problem is to investigate if it is possible to create an AI system to accurately distinguish various skin cancers.

Proposed Solution

Our proposed solution employs a basic deep-learning algorithm that accurately identifies skin cancer on diverse skin types and lesion variations.

For this project, we employed the HAM10000 dataset (“Human Against Machine with 10000 training images”), which is accessible on Kaggle/ Harvard Dataverse. Many scholars universally utilise this dataset as this dataset contains a collection of 10,015 standardised dermatoscopic images of different cancer types, lesion variations and lesions on different skin types. The lesion types included are melanoma, melanocytic nevi, basal cell carcinoma, actinic keratoses and intraepithelial carcinoma, benign keratosis-like lesions, dermatofibroma, and vascular lesions. This will be the training dataset for our 2 algorithms: support vector machine (SVM) and convolutional neural network (CNN).

Experiments

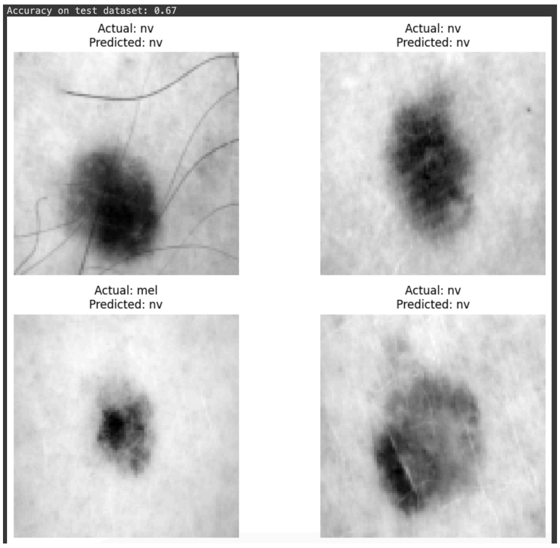
SVM

During experimentation, a Support Vector Machine (SVM) model was employed to classify skin lesions based on basic image details such as colours and shapes. To enhance the model's capability in detecting subtle texture and shape differences between various types of skin lesions, the Histogram of Oriented Gradients (HOG) technique was utilised. Initially, all images were standardised to a uniform size and converted into numerical lists, enabling the SVM to learn from overall picture patterns.

The approach was customised by integrating HOG features, computed from the grayscale images using an 8-orientation, 16x16 pixels per cell, and 1x1 cell per block configuration. This setup allowed for capturing finer details and textures within the images. Masks were applied to the images to isolate the lesions, further refining the feature extraction process. The dataset consisted of 10,015 standardised dermatoscopic images from the HAM10000 dataset. It was divided into training and testing sets, with 70000 images for training and 3000 for testing. By combining image data and HOG features, a comprehensive feature set was created for the SVM training.

The SVM model, trained with a radial basis function (RBF) kernel, achieved a test accuracy of 67%. The model was saved for future use, and predictions were generated for the test set. Visualisation of some of these predictions demonstrated the model's ability to distinguish between different skin lesion types. This experiment underscored the potential of using HOG features in conjunction with SVM for skin cancer classification, while also indicating areas for further improvement in accuracy.

```python
def load_skin_cancer_dataset(image_folder, csv_file, image_size=100, num_train=2000, num_test=500):
    images = []
    hog_features = []
    labels = []

    df = pd.read_csv(csv_file)

    # Shuffle the DataFrame
    df = df.sample(frac=1).reset_index(drop=True)

    train_count = 0
    test_count = 0

    for index, row in df.iterrows():
        image_path = os.path.join(image_folder, 'images', row.iloc[1] + '.jpg')
        mask_path = os.path.join(image_folder, 'masks', row.iloc[1] + '_segmentation.png')

        image = imread(image_path, as_gray=True)
        mask = imread(mask_path, as_gray=True)

        # Resize image and mask
        image = resize(image, (image_size, image_size))
        mask = resize(mask, (image_size, image_size))

        # Compute HOG features
        hog_feature = hog(image, orientations=8, pixels_per_cell=(16, 16), cells_per_block=(1, 1))

        if train_count < num_train:
            images.append(image.flatten())
            hog_features.append(hog_feature)
            labels.append(row.iloc[2])
            train_count += 1
        elif test_count < num_test:
            images.append(image.flatten())
            hog_features.append(hog_feature)
            labels.append(row.iloc[2])
            test_count += 1
        else:
            break

    images = np.array(images)
    hog_features = np.array(hog_features)
    labels = np.array(labels)

    return images, hog_features, labels, image_size
```



CNN Implementation

In order to implement a Convolutional Neural Network (CNN) to identify skin lesions from the HAM10000 dataset we chose to use transfer learning using pre trained CNN architectures. After testing multiple architectures such as VGG16, ResNet50 and Xception, an architecture known as DenseNet201 was chosen due to outperforming all others and its ability to achieve high accuracy on classification problems by efficiently utilising its parameters and promoting feature reuse through dense connections. DenseNet201 is an extension of the DenseNet architecture, characterised by its densely connected convolutional layers. The network comprises 201 layers, where each layer receives direct input from all preceding layers, enhancing gradient flow and feature propagation.

Before implementing the CNN model, data preparation had to be done. This included splitting the train, test and split datasets in a 70/15/15 split using sklearn’s libraries and using the ImageDataGenerator from Tensorflow’s Keras libraries to utilise data augmentation leading to less overfitting during training. This led to a total of 7010 images used for training, 1502 images for validation and 1503 images for testing.

To implement the DenseNet201 network, Python was used with the Tensorflow and Keras libraries. A sequential model was made using the DenseNet's inputs and outputs, followed by a Global Average Pooling 2D layer, a Dense Layer, a Dropout Layer and then the final prediction layer.

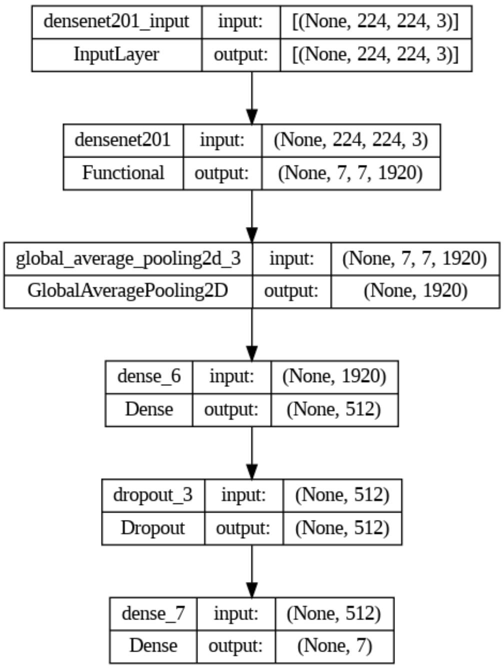


Figure. Implemented Model Summary.

The training phase was carried out using Google Colab's available A100 NVIDIA GPU that enabled us to leverage GPU acceleration to significantly speed up the training process. Initially, the model was trained using 5 epochs, a batch size of 32, the Adam optimiser with a learning rate of 0.001. The accuracy after training was 73.8% and the loss function was 0.7196. After this initial training took place and was evaluated, we proceeded with fine tuning the model. This was done by setting all layers in the DenseNet201 model to be trainable and implementing a learning rate reduction on plateau callback into the second training sequence. The model was then trained again, with the highest training accuracy reaching 97.05% with a loss function of 0.0852 however the validation accuracy peaked at 83.95%. The final testing accuracy for this model was 83.77% with a loss function of 0.6234. These results were able to demonstrate the models ability to perform well to unseen data and compared to other baseline models such as VGG16 (71.46%, (Khalil Aljohani & Turki Turki, 2022) with as ResNet50 (74.4%, Fraiwan & Esraa Faouri, 2022)

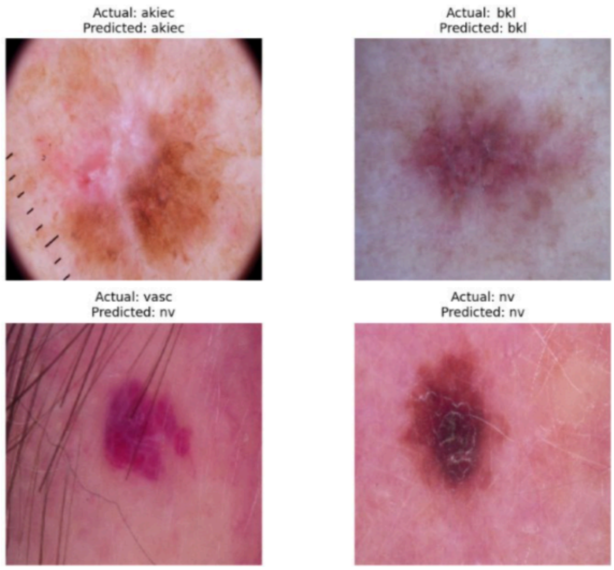


Figure. Results from DenseNet201 model.

Conclusion and Future Works

The application of AI in skin cancer detection has the potential to revolutionise healthcare, particularly in regions like Australia with some of the highest skin cancer incidence rates globally, at 8.7 times higher than the world average (Melanoma Institute Australia, 2022). Our study aimed to explore the feasibility of creating an AI system capable of accurately distinguishing various skin cancers. Utilising the HAM10000 dataset, we trained two algorithms—Support Vector Machine (SVM) and Convolutional Neural Network (CNN)—to identify skin lesions, there is significant room for improvement in both models' accuracy and robustness.

Future research will focus on refining these models to enhance their diagnostic accuracy. This includes optimising algorithm parameters, exploring more sophisticated models, and integrating multiple methodologies to better classify skin lesions. Additionally, we will investigate advanced image analysis techniques, such as image enhancement and region-focused analysis, to improve diagnostic precision.

Expanding our dataset to include more diverse skin types and leveraging real patient data, such as medical histories and test results, could further enhance our models' decision-making capabilities. Collaborating with dermatologists and other skin health experts will provide valuable insights to fine-tune our research.

Furthermore, developing a user-friendly application or website could democratise access to AI-powered skin cancer detection. This tool would allow users to upload photos of skin lesions and receive preliminary assessments, significantly improving early detection rates, especially in areas with limited access to dermatologists. By making these advancements, we can potentially reduce the progression and spread of skin cancer, ultimately improving public health outcomes.

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